

NETWORKS AND COMPLEXITY

Solution 17-3

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 17.3: Simple conservation law [Lvl. 2]

If you did the exercises of the previous chapters you might be able to guess the conservation law found in the dynamical system

$$\dot{x} = y \qquad \dot{y} = -x$$

Can you also find the conservation law mathematically?

Solution

This system featured in some previous exercises where we showed that the trajectories are a set of concentric circles around zero. So we might guess that the conservation law has the form

$$x^2 + y^2 = r^2 \tag{1}$$

where r is the constant radius of the circle. If we already know this it is tempting to convert to polar coordinates, which yields the equation $\dot{r} = 0$ which proves that the radius is conserved. However in this case a simpler approach that does not require prior knowledge of the solution is to use the method that was presented in this chapter. We write

$$\frac{dx}{dy} = \frac{\dot{x}}{\dot{y}} = -\frac{y}{x} \tag{2}$$

The equation

$$\frac{dx}{dy} = -\frac{y}{x} \tag{3}$$

can be separated into

$$x dx = -y dy \tag{4}$$

which we can integrate

$$\int x dx = - \int y dy \tag{5}$$

$$\frac{1}{2}x^2 = -\frac{1}{2}y^2 + C \tag{6}$$

which we can also write as

$$x^2 + y^2 = 2C \tag{7}$$

which is the same result as we guessed above, except that we named the conserved property $2C$ instead of r^2 , but that doesn't change anything.